

1. Graph and Graph Models

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∴ **Graph**: A graph $G = (V, E)$ → ^{vertex}edge consists of V , a non empty set of vertices (nodes) and E , a set of edges.

 **Multiple edge**: Many edge joining 2 same points

 **Loop**: edge joining the same points itself

pendant:- 1° degree

Isolated:- 0° degree

∴ **Simple graph**:- A graph in which each edge connects 2 different vertices and where no 2 edges connects the same pair of vertices is called a simple graph

• directed graph:- direction given

• undirected graph:- no direction given

In directed graph we justify:-

in-degree

out-degree

degree:- No. of lines attached to a points

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∴ **degree of a vertex**:- The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

The degree is denoted by $\deg(v)$ ^{name of vertex of which degree is to be removed}

∴ **Directed graph (Digraph)**:- A directed graph (V, E) consists of non empty set of vertices V and a set of directed edges (arcs) E . The directed edge associated with (u, v) [→] is said to start at u and end at v

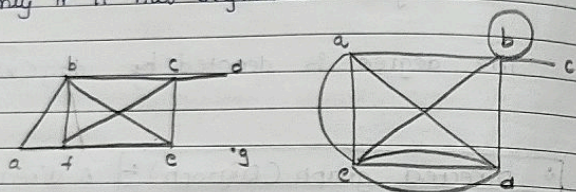
∴ **Simple directed graph**:- When a directed graph has no loops and no multiple directed edges, it is called a simple directed graph

∴ **Adjacent vertices**:- 2 vertices u and v are called adjacent if $(u, v) \in E$ if u and v are end points of an edge of G .

∴ Isolated vertex:- A vertex of degree 0 is called isolated vertex.

∴ Pendant vertex:- A vertex is pendant if and only if it has degree 1.

Homework:-



G

H

In the graph G and H find the degrees of each vertex, which vertex are pendant and which are isolated?

∴ In graph G:

• Vertex degree:

$$a = 2 \quad f = 4$$

$$b = 4 \quad g = 0$$

$$c = 4 \quad e = 3$$

$$d = 1$$

Isolated vertex = g

Pendant vertex = d

In graph H,

• Vertex degree:

$$a = 4 \quad e = 6$$

$$b = 6 \quad d = 5$$

$$c = 1$$

Isolated vertex = -

Pendant vertex = c

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∴ Handshaking theorem (lemma):

Let $G = (V, E)$ be an undirected graph with e edges then $2e = \sum_{v \in V} \deg(v)$

→ handshake

• Example:- How many edges are there in a graph with 10 vertices each of degree 6?

Ans

⇒ 10 vertices each have 6 degree

$$\therefore 2e = 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6$$

$$\therefore e = \frac{60}{2}$$

$$e = 30$$

Number of edges is = 30

The number of vertices of odd degree in a graph is always even.

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Proof: An undirected graph has an even number of vertices of odd degree.

odd degree the vertices even still,

Let V_1 and V_2 be the set of vertices of even degree and set of vertices of odd degree respectively in an undirected graph $G = (V, E)$ then $2e = \sum_{v \in V} \deg(v)$

$$= \underbrace{\sum_{v \in V_1} \deg(v)}_{\text{even}} + \underbrace{\sum_{v \in V_2} \deg(v)}_{\text{odd}} \rightarrow \text{by handshaking theorem}$$

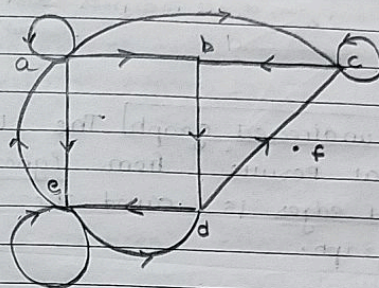
As degree of v_i [$\deg[v_i]$] is even for each v_i , $\sum \deg(v_i)$ is even.

also the L.H.S is even as it is $2e$.

Therefore, the last term on the R.H.S must be even, as all the terms in the sum of last term are odd, there must be even number of such term. Therefore, there are even number of vertices of odd degree.

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Q7 Find the in-degree and out-degree of each vertex in graph G.

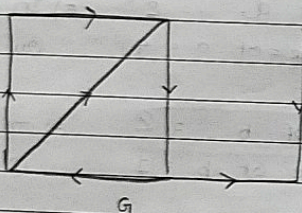


- 1) In-degree of d = 2
Out-degree of d = 2
- 2) In-degree of b = 2
Out-degree of b = 1
- 3) In-degree of c = 3
Out-degree of c = 2
- 4) In-degree of a = 2
Out-degree of a = 4
- 5) In-degree of e = 3
Out-degree of e = 3
- 6) In-degree of f = 0
Out-degree of f = 0

Let $G = (V, E)$ be a graph with directed edges then $\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$
 $\downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow$
 Sum of all in degrees Sum of all out degrees Total no. of edges

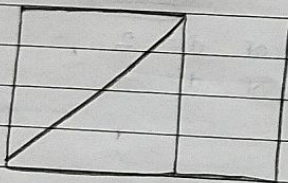
Undirected graph The undirected graph that results from ignoring directions of edges is called the undirected graph.

ex:-



G

The undirected graph of G is:-



Some special simple graphs:-

1] Complete graph : $[K]$

$\downarrow$
No. of vertices

→ degree of each vertex = $n-1$

→ $\sum \deg(v) = n(n-1)$

→ $|E| = \frac{n(n-1)}{2}$

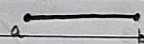
→ Each vertex is attached to every other vertex excluding itself

The complete graph on n vertices, denoted by K_n , is the simple graph that contains exactly one edge between each pair of distinct vertices.

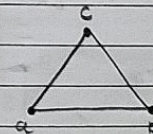
ex:- K_1

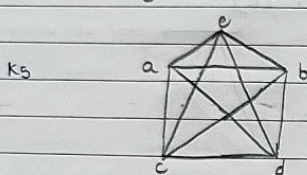
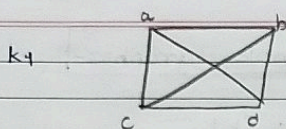
a

K_2



K_3





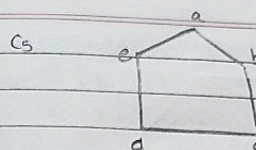
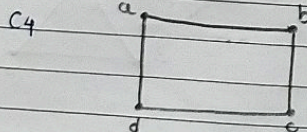
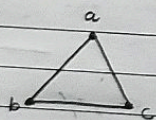
Note The no. of edges in K_n are $\frac{(n)(n-1)}{2}$

10) Cycle graph: (C_n)

→ Complete cycle
→ Starts only from C_3

The cycle (C_n) $[n \geq 3]$ consists of n vertices and the edges forming a cycle.

ex:- C_3



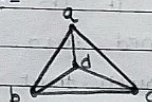
Note: There are n edges in C_n

5) Wheel graph $[W_n]$: To obtain W_n , with we draw C_n and add a vertex in the center and connect this new vertex to each of the n vertices in C_n .

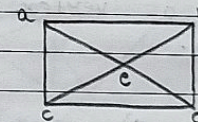
$W_n; n \geq 3$

for example:-

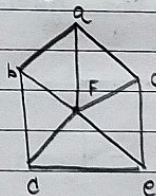
W_3 :



W_4 :



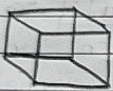
W_5 :



43] n-cube (Q_n) :

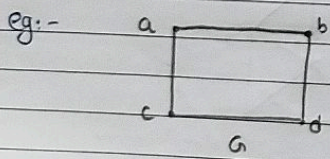
Q_1 

Q_2 

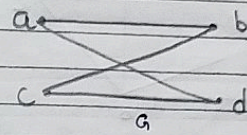
Q_3 

5] Bipartite graph : A simple graph is called bipartite if its vertex set V can be partitioned into 2 disjoint sets V_1 & V_2 such that every edge in G connects a vertex in V_1 to a vertex in V_2 .

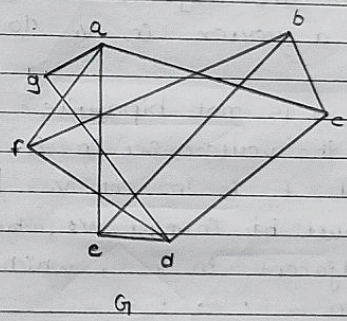
→ ^{not} adjacent in one group.

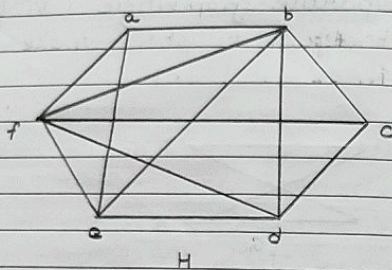


G is bipartite graph where $V_1 = [a, c]$
 $V_2 = [b, d]$



Q] Are the graphs G and H bipartite ? 28/Nov/2024





H

Graph G is bipartite. As, the vertex set can be split into 2 sets $V_1 = \{a, b, d\}$ and $V_2 = \{g, c, f, e\}$ so that no 2 vertices are adjacent in V_1 and no 2 vertices are adjacent in V_2 . Every edge joins a vertex in V_1 to a vertex in V_2 .

Graph H is not bipartite. As, we cannot split the vertex set in 2 disjoint sets as if f is in set V_1 then $\{a, b, c, d, e\}$ all must be in set V_2 , but b and c are adjacent in V_2 , which should not happen.

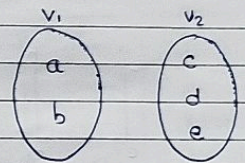
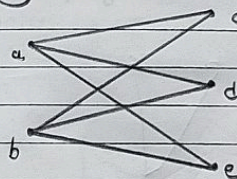
6. Complete bipartite graph:- $[K_{m,n}]$

each vertex in V_1 should make edge with each in V_2

A graph that has its vertex set partitioned into 2 sets and there is an edge between 2 vertices if and only if one vertex is in the first set and the other vertex is in 2nd set.

ex:-

$K_{2,3}$: no. of vertex in each V_1, V_2



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7. Regular graph:- A simple graph is called regular if every vertex of the graph has the same degree. It is called n-regular if every vertex has degree n.

eg:-



1- regular



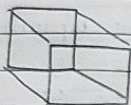
2- regular



2- regular



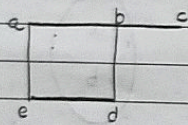
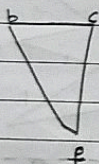
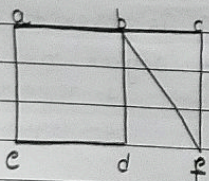
5-regular



3-regular

* Union of graphs:- The union of graphs $G_1 = [V_1, E_1]$ and $G_2 = [V_2, E_2]$ is the simple graph with the vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$.

ex:- find the union of the graphs G_1 & G_2

 G_1  G_2  $G_1 \cup G_2$

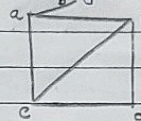
undirected graph $\rightarrow$ symmetric matrix

directed graph $\rightarrow$ not

∴ Adjacency list and Adjacency matrix:-

Q1 Find the adjacency list of the following graph

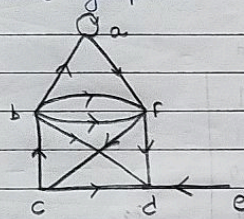
Ans $\Rightarrow$

 G

Vertex	Adjacent vertices
a	b, c, e
b	a
c	a, d, e
d	e, c
e	a, c, d

Q2 Find the adjacency list of the following

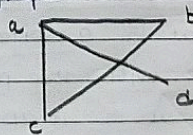
~~undirected~~ graph:-

 G

Initial-vertices	Termince Adjacent ver
a	f, a
b	a, f, d
c	b, d
d	
e	d
f	c, d

Q3 find the adjacency matrix of the following graph:-

$\Rightarrow$



Symmetric Matrix $\Rightarrow \begin{matrix} a_{11} = a_{11} \\ a_{12} = a_{21} \\ a_{23} = a_{32} \end{matrix}$ } same.

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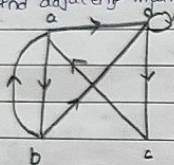
No. of vertices: being these

$\Rightarrow$

	a	b	c	d
a	0	1	1	1
b	1	0	1	0
c	1	1	0	0
d	1	0	0	0

$\rightarrow$ degree of a $= 3^0 = 1+1+1$

Q1) Find adjacency matrix:-



G

	a	b	c	d
a	0	1	0	1
b	1	0	0	1
c	1	0	0	0
d	0	0	1	1

Isomorphism of graph :- Simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there is a one to one correspondence between the vertices G_1 & vertices of G_2 , that is a bijective function f from V_1 to V_2 with the property that a & b are adjacent in G_1 if and only if $f(a)$ & $f(b)$ are adjacent in G_2 . Such a function f is called isomorphism.

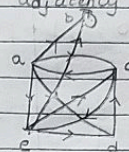
H.W Q1) Represent the following graph using adjacency matrix:-



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Q2) Find adjacency list of the following graph:-



Note:- Out going graphs are considered.

Ans Q1]

	a	b	c	d
a	0	3	0	2
b	3	0	1	1
c	0	1	1	2
d	2	1	2	0

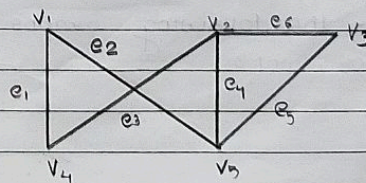
Ans Q2]

Initial - vertices	Terminal vertices
a	c, d, b, e
b	b
c	a, e
d	c
e	d, c, b

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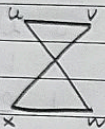
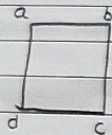
Incidence Matrices :-

Ex:- Find the incidence matrix of the following graph:-



Ans →

	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	0	1
v_3	0	0	0	0	1	1
v_4	1	0	1	0	0	0
v_5	0	1	0	1	1	0

isomorphism
x = boxed $a \leftrightarrow u$ $b \leftrightarrow v$ $c \leftrightarrow w$ $d \leftrightarrow x$ graph invariant → No. of edge
[preservative] → No. of vertex

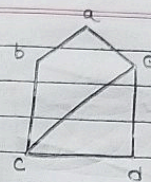
for getting → Vertices degree

isomorphism betⁿ → No. of edges betⁿ particular graph.

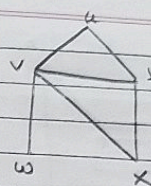
2 graph

HK

HK

Check whether the following graphs are
isomorphic or not:-

G

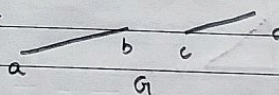


H

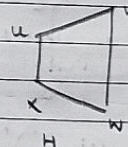
The degree of vertex v in graph H is 4. But no. vertex in graph G has degree 4. So we do not get a correspondence for vertex v hence graph G and H are not isomorphic.

Also, the vertex w in H is a pendent vertex but graph G has no pendent vertex. Hence G and H are not isomorphic graph.

Ex 2:-



G



H

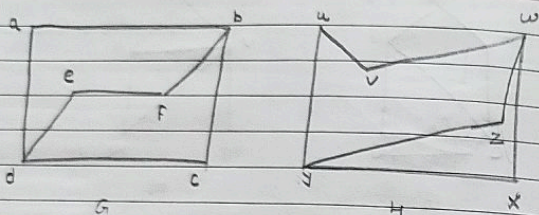
There are only 2 edges in graph G and 4 edges in graph H hence the number of edges are different in both the graph. Also, each vertex in graph H has degree 2 whereas, each vertex in graph G is pendent vertex. Hence G & H graph are not isomorph.

Adjacency Matrix - vertices to vertices
Incident Matrix - vertices to edge

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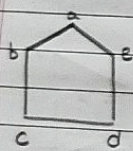
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Ex 3]

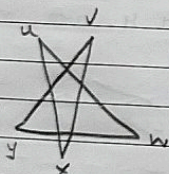


The graph G and H are isomorphic, the corre

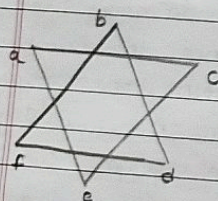
$a \leftrightarrow x$
 $b \leftrightarrow w$
 $c \leftrightarrow z$
 $d \leftrightarrow y$
 $e \leftrightarrow u$
 $f \leftrightarrow v$



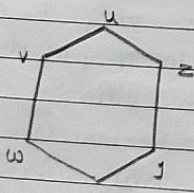
G



H



G



H

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Q4] Ans - The graph G and H are isomorphic, the corresponding edges are as follows:-

$a \leftrightarrow y$
 $b \leftrightarrow v$
 $c \leftrightarrow x$
 $d \leftrightarrow u$
 $e \leftrightarrow w$

Q5] Ans - In graph G and H, we do not get any corresponding edges. If we consider vertex a and c adjacent to c and e and c and e also have an edge between them. But in graph H, no 2 vertices adjacent to a particular vertex have a edge in between them. Hence, graph G and H are not isomorphic.

Revised Ans

Q5] Ans - A graphs G and H, are not isomorphic as we do not get a correspondence betⁿ the vertices of G and vertices of H. The vertex b is adjacent to f & d, d is adjacent to b & f, f is adjacent to b & d but neither of the 3 vertices are adjacent to the remaining 3 vertices whereas in graph H

all the six vertices form a circuit.
Hence, graph G & H are not isomorphic.

Connectivity:-

[walk]

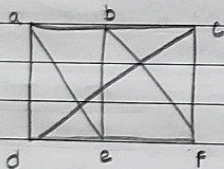
i) Path:- A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along the edges of the graph.

simple path

ii) Trail:- Trail is a walk that has no repeated edge. It is also called simple path.

closed path

iii) Circuit:- A path of length greater than zero that begins and ends at the same vertex is called the circuit or cycle.



Q1] find a walk of lengths 2, 3, 4 & 5 from a to f & c to a.

Q2] find a trail from a to b of length 5
find a circuit from b to d.

all trail is walk

all walk is not trail

Q1] a to f

$a \rightarrow b \rightarrow c \rightarrow f$

$a \rightarrow d \rightarrow c \rightarrow e \rightarrow b \rightarrow f$

$a \rightarrow d \rightarrow c \rightarrow b \rightarrow e \rightarrow f$

c to a $c \rightarrow b \rightarrow a$

$c \rightarrow f \rightarrow b \rightarrow a$

$c \rightarrow f \rightarrow b \rightarrow e \rightarrow a$

$c \rightarrow b \rightarrow f \rightarrow e \rightarrow d \rightarrow a \rightarrow a$

trail c to a = $c \rightarrow b \rightarrow a$

$c \rightarrow d \rightarrow a$

$c \rightarrow f \rightarrow e \rightarrow d \rightarrow a$

$c \rightarrow f \rightarrow e \rightarrow d \rightarrow b \rightarrow a$

Q1] a to f

2nd length $\Rightarrow a \rightarrow e \rightarrow f$

3rd length $\Rightarrow a \rightarrow b \rightarrow c \rightarrow f$

5th length $\Rightarrow a \rightarrow d \rightarrow c \rightarrow b \rightarrow c \rightarrow f$

4th length $\Rightarrow a \rightarrow e \rightarrow b \rightarrow c \rightarrow f$

c to a

2nd length $\Rightarrow c \rightarrow a \rightarrow f$

3rd length $\Rightarrow c \rightarrow b \rightarrow e \rightarrow a$

4th length $\Rightarrow c \rightarrow f \rightarrow e \rightarrow b \rightarrow a$

5th length $\Rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow b \rightarrow c$

1 component $\rightarrow$ connected
 More than 1 $\rightarrow$ disconnected

vertex removed $\rightarrow$ then all edge attached to it also get removed

if edge removed only

edge is cut

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Q2] trail from a to b $\rightarrow a \rightarrow d \rightarrow e \rightarrow f \rightarrow c \rightarrow b$ [length 5]
 circuit from b to b $\rightarrow$

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Q3] Define connectivity :-

$\Rightarrow$ An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph.

o Theorem :-

There is a simple path [trail] betn every pair of distinct vertices of a connected undirected graph.

o Connected component :-

A connected component of a graph G is a connected sub graph of G that is not a proper sub graph of another connected subgraph of G .

o Cut vertex :-

Cut vertex is a vertex whose removal produces a graph with more connected components than in the original graph.

o Cut edge (bridge) :-

An edge whose removal produces a graph with more connected

strongly connected :- every vertex should have incoming and outgoing in directed graph.

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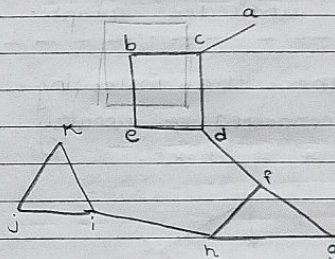
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components than in the original graph with more connected components than in the original graph is called cut edge.

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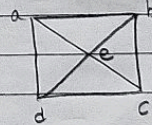
Q4] Find the cut edge and cut vertex in the following graph :-

1]



Cut vertices in the graph are :- f, h, i
 Cut edge in the graph are :- $d-f, h-i$.

2]



Cut vertices :- No
 Cut edge :- No

• weak conn. :- 1 piece

• strong conn :- in case if out come both from vertex

if strongly connected then weakly connected

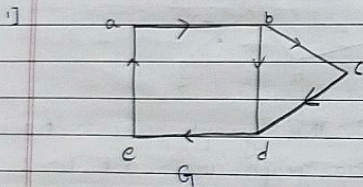
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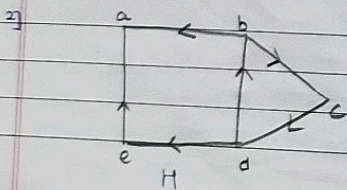
◦ Strongly connected :- A directed graph is strongly connected if there is a path from a to b and from b to a whenever a & b are vertices in the graph.

◦ Weakly connected :- A directed graph is weakly connected if there is a path between every 2 vertices in the undirected graph.

Q] Check whether the following graphs are strongly connected or weakly connected or both or neither :-



Ans $\Rightarrow$ Strongly connected as well as weakly connected

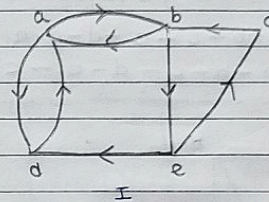


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Ans $\Rightarrow$ The graph H is weakly connected because its undirected graph is connected. H is not strongly connected as there is no path from a to b .

3]



The graph I is weakly connected as undirected graph is

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◦ Euler and Hamilton path

◦ Euler paths :- A path containing every edge of G is called Euler path in G .

◦ Euler circuit :- An Euler circuit in a graph G is a simple circuit containing every edge of G .

eg:-

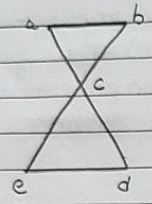
Q] Find the Euler circuit in the following graphs along with Euler path.

Euler path:- all edge covered + no repetition of edge

Euler path :- $||$ + same start & end point

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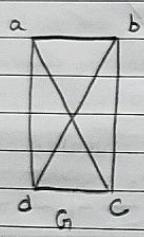
1)



Euler path:- $a-b-c-e-d-c-a$

Euler circuit:- $a-b-c-d-e-c-a$

2)

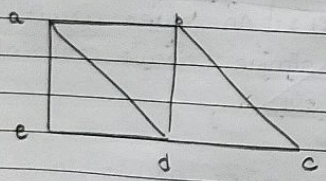


Euler path:- Does not exist

Euler circuit:- Does not exist

The graph G has no Euler path and no Euler circuit

3)



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Ans:-

Euler path:- $a-b-c-d-e$

Euler ~~circuit~~ ^{path 2}:- $a-e-d-c-b-a-d-b$

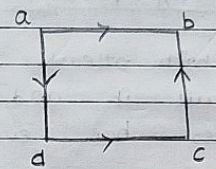
The graph H has Euler paths are as follows

[1] $a-d-b-c-d-e$

[2] $a-e-d-c-b-a-d-b$

but there is no Euler circuit.

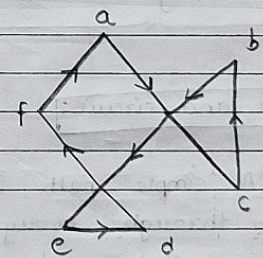
4)



Ans:-

The graph has no Euler path as well as no Euler circuit.

5)



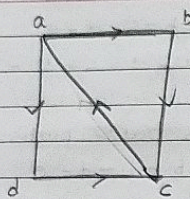
Ans:-

The Euler path and circuit in the graph is $a-c-b-e-d-f-a$

Euler:- No repetition of edge but vertex
 Hamilton:- No repetition of vertices but edge

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5)



Ans $\Rightarrow$ The graph has no Euler circuit,
 Euler path in the graph is $a-b-c-d-a$

Theorem: A connected graph with atleast 2 vertices
 has an ~~atleast~~ ^{Euler} circuit if and only if
 each of its vertices has even degree

Theorem: A connected graph has an Euler path
 but not an Euler circuit if and only
 if it has exactly 2 vertices of odd degree

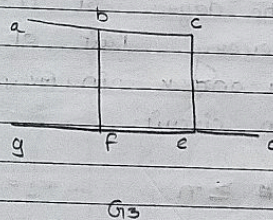
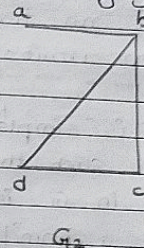
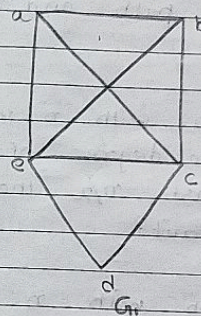
* Hamilton path and circuits:

Hamilton path: A simple path in a graph
 G that passes through every vertex
 exactly once is called a Hamilton path.

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Hamilton circuit: A simple circuit in a graph
 G that passes through every vertex
 exactly once is called a Hamilton circuit
 for ex:-

Check whether the following graphs



In the graph G_1 , Hamilton path and circuit
 both exists Hamilton path is $a-b-c-d-e$
 Hamilton circuit is $a-b-c-d-e$

G_2 :

There is no hamilton circuit

Hamilton path in G_2 is $a-b-c-d$

G_3 :

There is no hamilton path and no hamilton circuit.

Theorem: IF G is simple graph with n vertices $n \geq 3$ such that the degree of every vertex in G is atleast $n/2$ then G has a hamilton circuit.

Theorem: IF G is a simple graph with n vertices, $n \geq 3$ such that the degree of u + degree of $v \geq n$ for every pair of non adjacent vertices u and v in G , then G has a hamilton circuit.

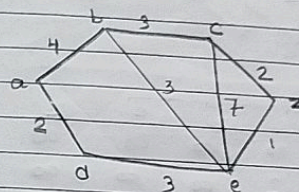
$$\deg(u) + \deg(v) \geq n \quad \text{in deg = out degree}$$

Shortest path problems:-

Weighted graphs:- The graphs that have a number assigned to each edge are called weighted graphs

Shortest path algorithm:-

Using Dijkstra's Algorithm find the shortest path from A to Z.

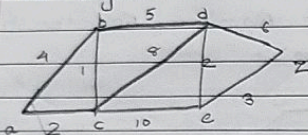


Dijkstra's Algorithm are as follows:-

Lengths and paths form:-

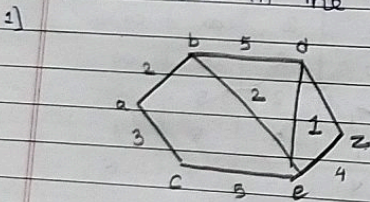
	a	b	c	d	e	z
a	0	∞	∞	∞	∞	∞
a,d	0	4	∞	2	∞	∞
a,d,b	0	4	∞	2	5	∞
a,d,b,c	0	4	7	2	5	9
a,b,d,c,e	0	4	7	2	5	6

5] Find the shortest path from A to Z using the algorithm:-

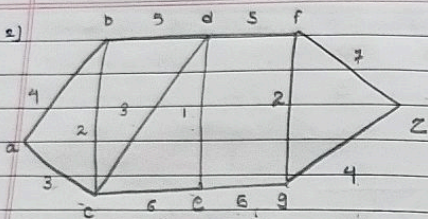


	a	b	c	d	e	z
a	0	∞	∞	∞	∞	∞
a, c	0	4	2	∞	∞	∞
a, c, b	0	3	2	10	12	∞
a, c, b, d	0	3	2	8	12	∞
a, c, b, d, e	0	3	2	8	10	13

9] Find the length of the shortest path from A to Z in the following graph:-



	a	b	c	d	e	z
a	0	∞	∞	∞	∞	∞
a, c	0	2	13			

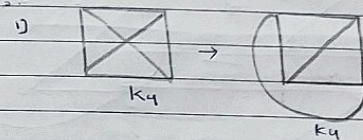


$A = c - d - e - g - z$
 $A = c - e - g - z$

	a	b	c	d	e	f	g	z
a	0	∞	∞	∞	∞	∞	∞	∞
b	0	4	3	6	9	0	0	0
c	0	4	3	6	9	0	0	0
d	0	4	3	6	9	∞	∞	∞
e	0	4	3	6	7	11	∞	∞
f	0	4	3	6	7	11	11	∞
g	0	4	3	6	7	11	13	18
z	0	4	3	6	7	11	13	16

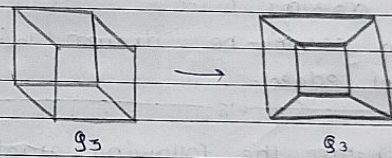
Planar Graphs:- A graph is called planar if it can be drawn in the plane without any edges crossing such a drawing is called a planar representation of the graph.

ex:-



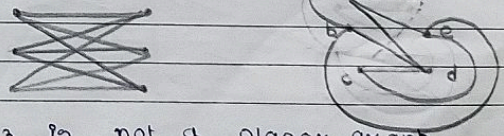
Hence K_4 is plane graph.

2)



Q_3 is planar.

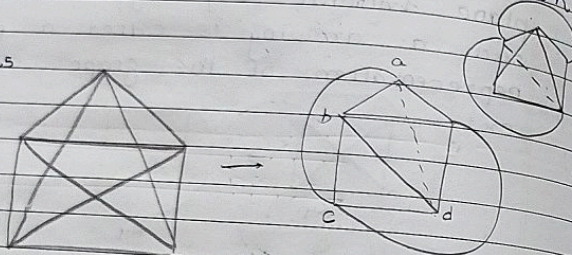
3)



$K_{3,3}$ is not a planar graph. As it cannot be drawn in the plane without crossing.

We cannot draw the edge cd without crossing of edges hence $K_{3,3}$ is non planar

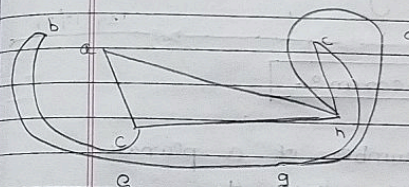
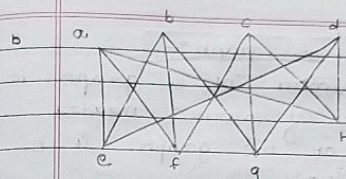
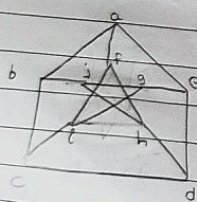
4] K_5



K_5 is not a planar graph because it cannot be drawn in the plane without crossing. Edge ad cannot be drawn without crossing of edges.

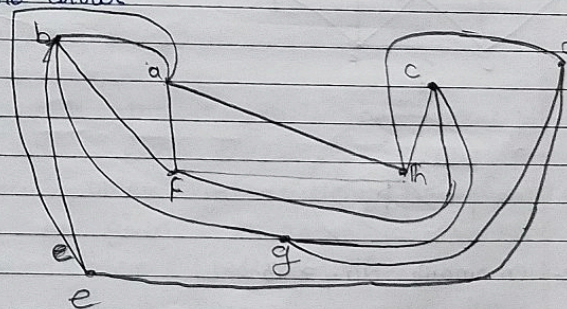
9] Check whether the following graphs are planar or not:-

a



Graph colouring :- A colouring of simple graph is the assignment of a colour to each vertex of a graph so that no 2 adjacent vertices are assigned the same colour

b



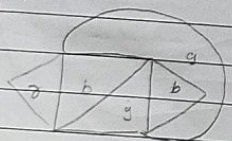
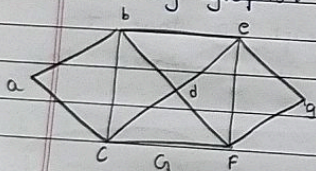
Chromatic Number of graph:-

The chromatic Number of a graph is the least number of colours needed for a colouring of the graph it is denoted by $\chi(G)$ or χ .

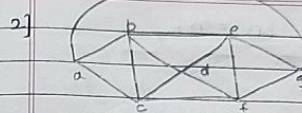
The 4 colour theorem:-

Theorem:- The chromatic number of a planar graph is no greater than 4.

Ex:- What are the chromatic numbers of the following graphs:-

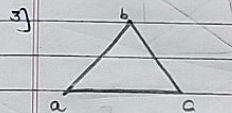


Ans:- Chromatic No. = 3.

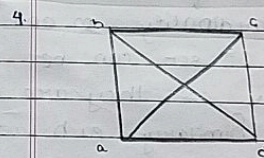


a → Red
b → Green
c → Blue

Ans → Chromatic no. = 4.



Ans → Chromatic no. = 3.



Ans → Chromatic No. = 4

Q] What is the chromatic no. of K_n :-

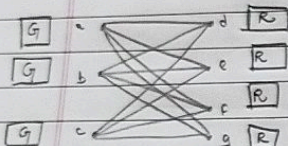
⇒ In K_n , every vertex is adjacent to every other vertex. Hence, No 2 vertices can be given the same colour, hence each vertex needs a different colour. There are n vertices in K_n . Hence

Total elements in sets } cardinality

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chromatic number of $K_n = n$.

1] $K_{3,4}$



$R \rightarrow$ Red
 $G \rightarrow$ Green

Q] What is chromatic no of $K_{m,n}$ [complete bipartite graph?

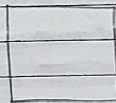
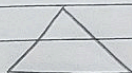
$\Rightarrow$ We can separate vertices of $K_{m,n}$ into 2 distinct sets with cardinality m and n . Each vertex in the 1st set can be given the same colour as they are not adjacent to each other. Similarly, each vertex in the 2nd set can be given the same 2nd colour. Hence $K_{m,n}$ can be coloured by using only 2 colours. Therefore, $\chi(K)$ = 2

Q] What is the chromatic number of C_n :-

$\Rightarrow$ If n is even, then we can assign 2 colours alternately hence $\chi(C_n)$ = 2 if n is even

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If n is odd, we can assign 2 colours alternately and the last vertex needs to be assigned a different colour hence $\chi(C_n) = 3$ if n is odd.



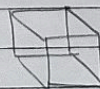
$\therefore W_n$:-

odd $\Rightarrow 3$
even $\Rightarrow 2$

$\therefore \chi(C_n) \Rightarrow$ 2 in every case.

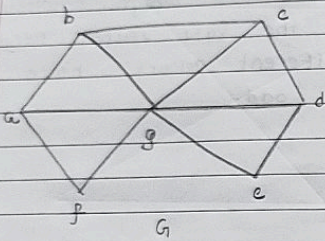


$\chi_2 \rightarrow 2$



$\chi_3 \rightarrow 2$

H.W:-



Q) Find chromatic no. of the given number.

1 Introduction to graph
Practical 1

Q1] Undirected graph :-

 $V = \{a, b, c, d, e\}$, $E = \{(a,b), (a,d), (b,c), (d,b), (c,d), (e,a), (e,b), (e,c)\}$
⇒ input + output

load(" graphs");

a: 1;

b: 2;

c: 3;

d: 4;

e: 5;

 $G := \text{create_graph}([a,b,c,d,e], [(a,b), (a,d), (b,c), (d,b), (c,d), (e,a), (e,b), (e,c)]);$
 $c := / \text{maxima} - 5.45.1 / \text{share} / \text{maxima} / 5.45.1 /$
 $\text{share} / \text{graphs} / \text{graphs} . \text{mac}$

1

2

3

4

5

Graph (5 vertices, 8 edges)

draw_graph(G, show_id = true);